LUXEMBOURG DIGITAL BANK · ACCOUNT-TAKEOVER FRAUD DETECTOR

TrustAge

A criminal can steal a password. Not time.



Charalampos Thanasis



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Every takeover has to add something new

Purpose: stop takeover before the money leaves, explain every decision, work from day one.

AFTER A TAKEOVER

DEVICE	PHONE, E-MAIL	SESSION	PAYEE	DESTINATION
2 min	**2 days**	**jumped**	**10 min**	**3 strangers**
first seen just now	changed just before	new network mid-way	never paid before	pay it within 72 h
R2	R1	R2c	R3 · R5	R6
Phishing, fake clone site	SIM swap, e-mail takeover	Real-time relay, stolen cookie	Fake bank caller, "safe account"	Money mule

€2.2 bn
credit-transfer fraud, EEA 2024

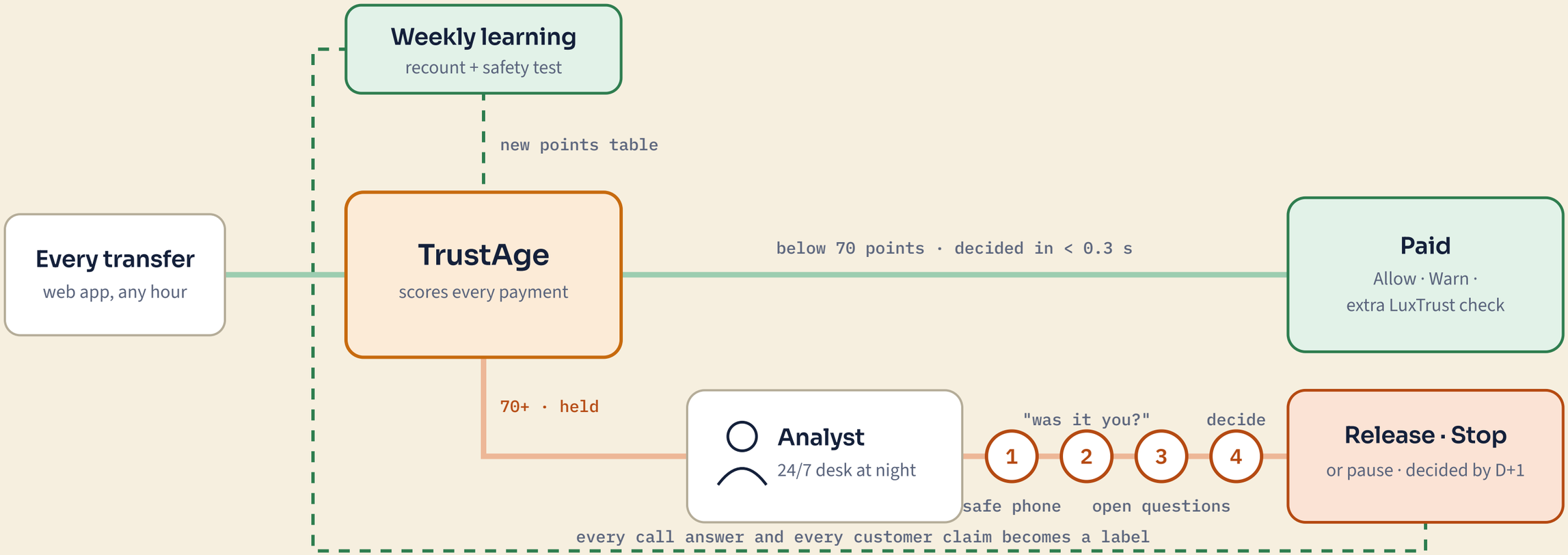
10 s
for an instant payment to leave

0
days of history at LDB

ASSUMPTIONS

- Trust starts at verified onboarding
- Strong authentication already on every transfer
- Plan: 1 fraud in 20,000
- Terms allow a safety hold

The machine decides most. People decide holds.

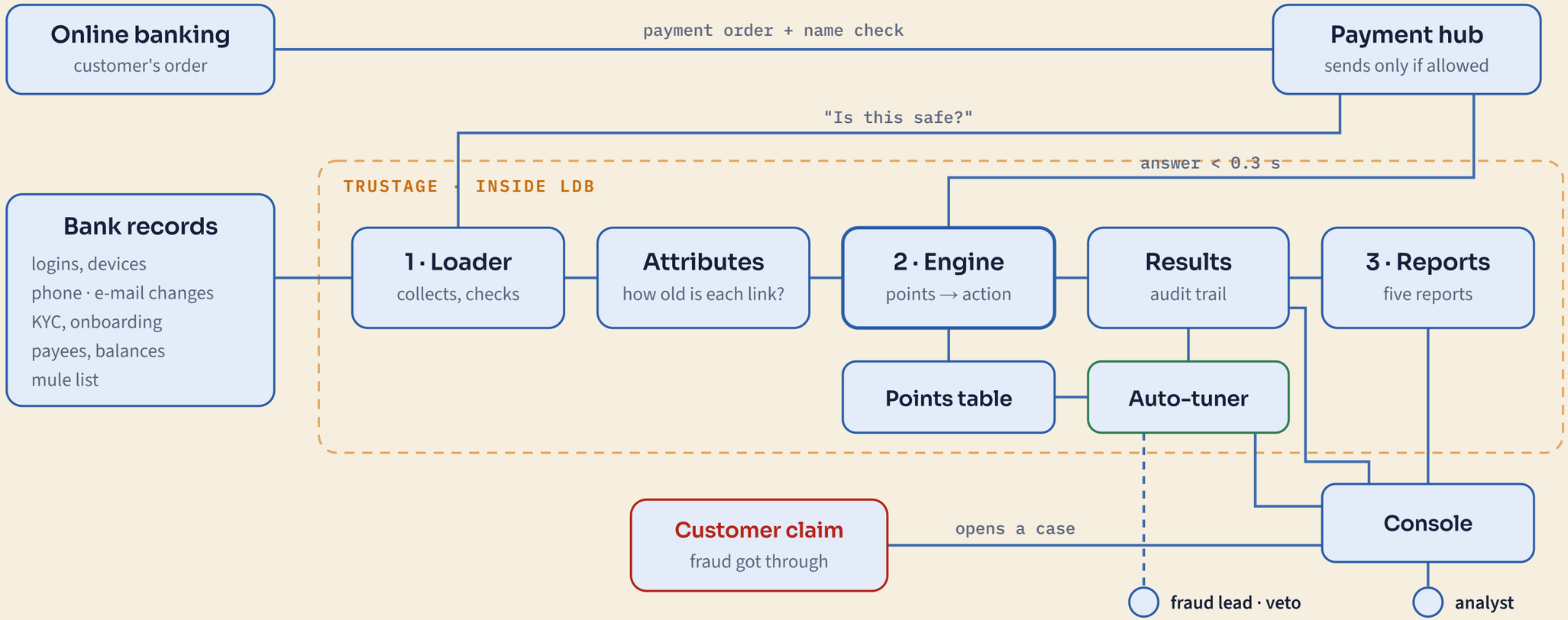


Fraud lead reads the weekly log, can veto

Risk · 2nd line, monthly report

Audit · 3rd line, every version stored

Three parts inside LDB, one answer in 0.3 s

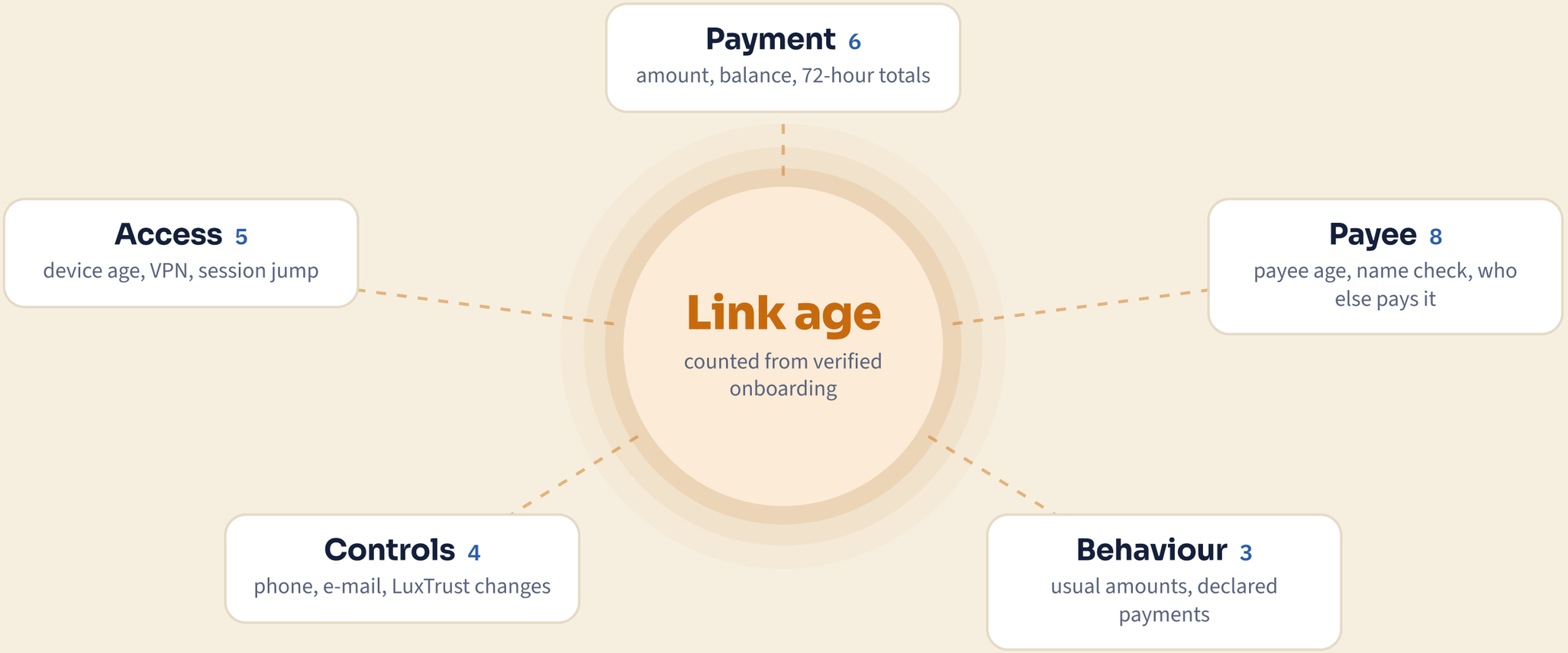


Instant payment held? Refused within 10 s, then continues as a standard transfer, decided by D+1.

TrustAge down? Trusted payees still pass; €1,000+ to new payees waits.

DATA FEATURES USED

34 attributes, and none needs history



26 essential + 8 supplementary · a missing essential field counts as risky · device IDs hashed

Every point comes from one formula

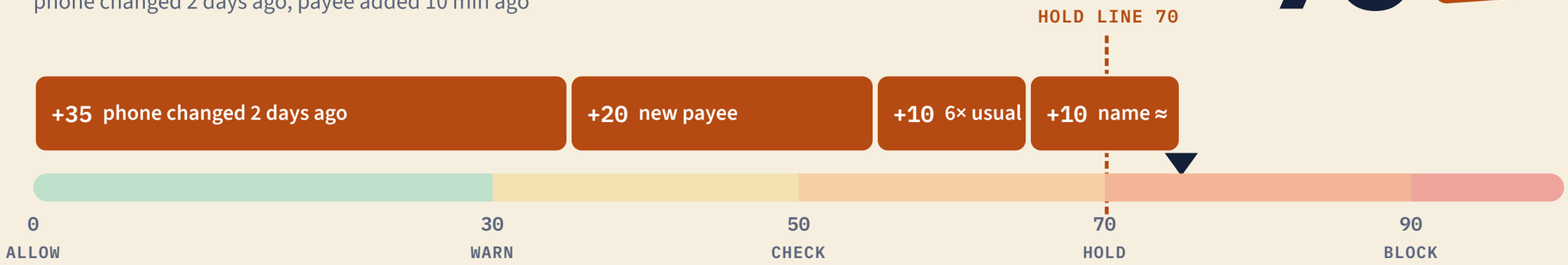
$$\text{points} = 7 \times \log_2 \frac{F}{G}$$

F share of frauds with the signal · **G** share of genuine payments with it

Takeover · €4,800 instant

phone changed 2 days ago, payee added 10 min ago

75 HOLD



R1	R2	R3	R4	R5	R6	R7	R8	M1–M5	01–04
control change	new device, network	fresh payee	name mismatch	draining	mule account	coached session	test, dormant	good signs	fixed overrides

Alert priority: **P1** within 15 min, 24/7 · **P2** within 1 hour · **P3** next business day · a case takes its highest signal's priority

One report for each person and decision

Alert card — analyst · each hold | **Queue** — analyst · live | **Campaign board** — fraud lead · daily | **What it learned** — lead, risk · monthly | **KPI page** — management · CSSF

Case A-0042 · €4,800 instant

P1 · HELD · 75

11:24 left

WHY, IN ONE SENTENCE

Phone number changed 2 days ago, then €4,800 (6× her usual) to a payee added 10 minutes ago whose name only partly matches.

R1	Phone changed 24–72 h ago	+35
R3	Payee added 10 min ago	+20
R3	Amount 6× her usual	+10
R4	Name only a close match	+10
	Total · hold line 70	75

LINK AGES

Device	Anchored · 92 days
Phone	Changed 2 days ago
Payee	Added 10 min ago

CALL-BACK

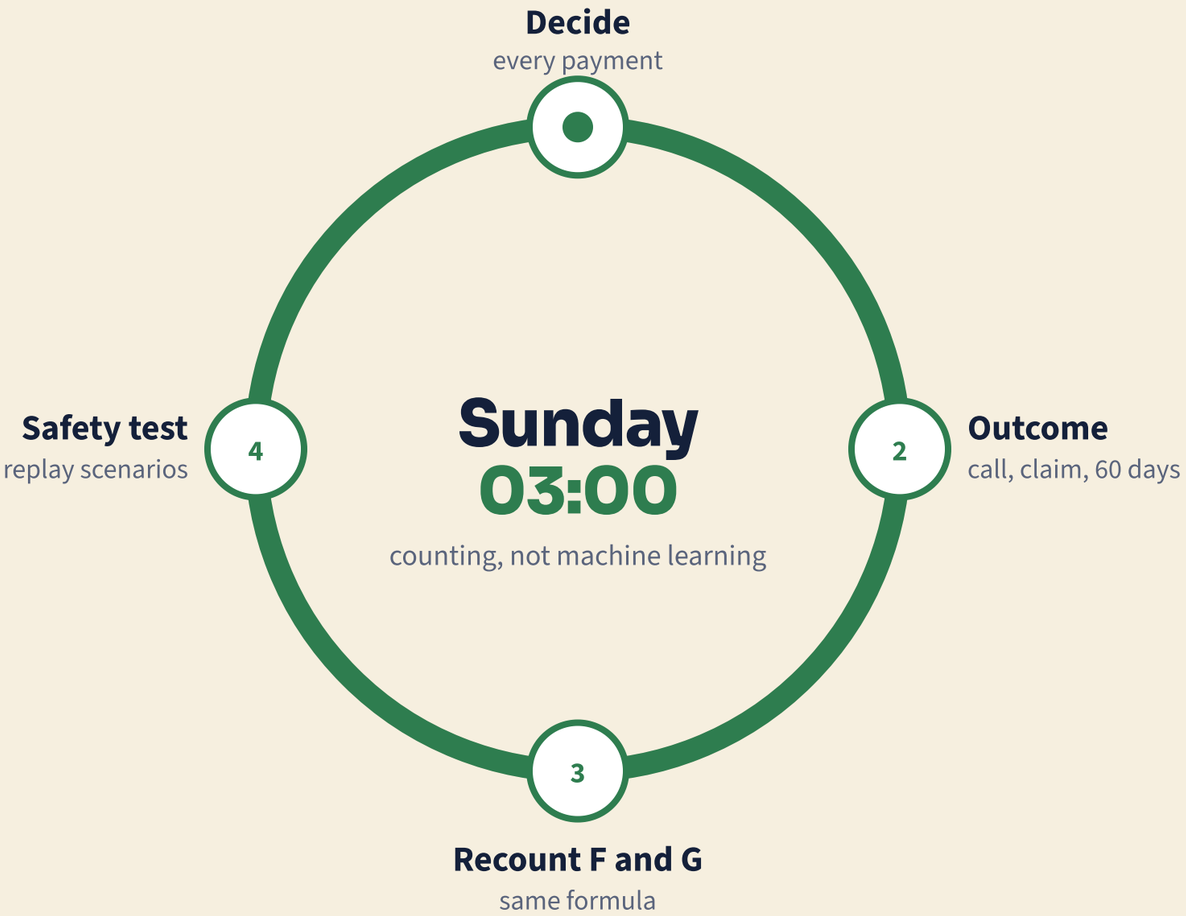
1 · Call the **safe phone** from onboarding, never the new number

2 · "Did you make this payment?"

Release | Stop + block | Pause to D+1

Report mockups · example numbers, month 4 · click a tab or press →

Every Sunday it recounts the evidence



LOG · MONTH 4 · TABLE v1.4

R3 fresh payee +20 → +25

21 of 25 confirmed frauds had one;
only 5.5% of genuine payments do.

✓ every library scenario still reaches its action

±5 points a week at most

Floors on core rules

Overrides change only with four eyes

New rule starts at 0 points

Veto and one-click rollback

TESTING · LIMITS · FIT

Strong on takeovers, honest about the rest

Desk check: one scenario per attack route plus five honest payments, scored with the launch points table

9/12

fraud scenarios stopped

BLOCKED OR HELD

2

slowed by the check and a 1-hour release

F8 · F11

1

missed: coached victim, €900, own device

F12

5/5

honest payments paid

INCL. THE €320,000 FLAT

Coached victim, own device

Slowed by a check question and a 1-hour release, not stopped.

Patient criminal who waits

Scores lower, so warnings also go to the old phone and e-mail for 30 days.

Mule used only once

Needs mule data shared between banks.

PSD2 + SCA RTS · Instant Payments Reg. · VoP · GDPR Art. 22 · AI Act: not high-risk · DORA

≈ €0.48 per transfer

Thank you

TrustAge measures the one thing a criminal cannot fake: the age of each link.



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QUESTIONS